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A revised magnetostratigraphy for the Mid-Proterozoic Gardar lava succession, South Greenland

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ABSTRACT

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New palaeomagnetic results from the ca. 1300 Ma Gardar lava succession within the Eriksfjord Formation of South Greenland are presented. This study incorporates stepwise thermal demagnetisation of 256 samples from 66 lava flows from all three lava groups in this formation, accompanied by principal component analysis. The results identify two complete reversals (N → R → N) in the upper lava group and two (R → N and N → R) in the lower group, with a possible palaeoexcursion in the lower group. A minimum of two further reversals are predicted in the time interval represented by the sandstone units intercalated with the lava flows. Both complete reversals in the upper group exhibit inclination asymmetries of 30°. This degree of asymmetry is satisfied by a geomagnetic field model equivalent to a dipole axially offset by 570 km. However, a more complex source is not precluded by the present evidence. Four new pole positions are calculated for the Gardar lavas, two for the upper lava group and one each for the middle and lower groups. The middle lava group shows an anomalous mean direction and pole position, which is unrepresentative of the remainder of the lava succession and is thought to record a virtual geomagnetic pole representative of a transitional field direction. The presence of a non-axial geocentric dipole field means that the use of the Laurentian apparent polar wander path for dating the new poles is not valid.

1. Introduction

The definition of a magnetic stratigraphy for the Mid-Proterozoic (ca. 1300 Ma) Gardar lavas within the Eriksfjord Formation of South Greenland was the subject of a previous study by Piper (1977). That study resolved stable remanence directions chiefly by alternating field (A.F.) demagnetisation and identified five polarity reversals throughout the lower two thirds of the ca. 1500 m thick lava stratigraphy. Four of these reversals, within the upper lava group, showed an asymmetry of ca. 30° in inclination which suggested that the geomagnetic field source at the time that these lavas were erupted, may have been more complex than an axial geocentric dipole.

The present study has aimed to investigate the significance of this reversal asymmetry by resolving the detailed component structure in the lavas using thermal demagnetisation and modern techniques of component analysis (Kirschvink, 1980). This study also extends the section sampled by Piper (1977) to cover the entire 1500 m of lava stratigraphy, with the hope of identifying any further reversals in the previously unsampled upper 200 m of the lava pile. It has also aimed to clarify the significance of anomalous results obtained by Piper, mainly in the middle of the three lava groups.

2. Geology and sampling

The Gardar igneous province of South Greenland records an approximately 200 Ma period of persistent crustal movements and alkaline magmatism. It is located chiefly within the granitic

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TABLE 1

Simplified geological succession within the Gardar province

Period	Rock type	Locality	Rb–Sr age [a]
Late Gardar	ENE dykes (dolerite,trachyte, lamprophyre, phonolite)		
	Intrusive complexes	Igdlerfigssalik	1154 ± 15
		Ilímaussaq	1143 ± 21
		South Qôroq	1160 ± 8
Mid Gardar	WNW dykes (dolerite)		
	Intrusive complexes	Kûngnât	1219 ± 17
		Ivigtut	1222 ± 25
Early Gardar	Intrusive complexes	North Qôroq	1291 ± 61
		Motzfeldt	1291 ± 31
		Gronnedal-Ika	1299 ± 17
	Eriksfjord Formation	Ilímaussaq peninsula	

[a] Isotopic ages from Blaxland et al. (1978), corrected using formulae from Steiger and Jäger (1977).

basement of the 1850–1600 Ma Ketilidian mobile belt. Extensive isotopic dating of the Gardar intrusive rocks (Blaxland et al., 1978) has identified three tectono-magmatic events in the evolution of the Gardar province, each showing characteristic features of Neogene continental rift environments (Upton and Blundell, 1978). Table 1 lists the chronology of the Gardar cycle. The earliest

TABLE 2

Stratigraphical details of the Eriksfjord Formation

Group name	Petrological description	Environment
Ilímaussaq group (upper lavas)	Aphyric and feldspar–phyric basalts and trachytes.	Tectonically quiescent; quasi-horizontal plains.
Nunasarnaq group (upper sandstones)	Massive quartzitic aeolian sandstone.	Desert.
Ulukasik group (middle lavas)	Aphyric and feldspar–phyric basalts. Cavities filled with wind blown sand.	Tectonically quiescent; quasi-horizontal arid plains.
Naujarssuit group (middle sandsones)	Medium-fine grained grey quartzitic sandstones.	Fluviatile.
Mussartût group (lower lavas)	Alkaline ultramafics initially, but glassy basalts later. Interleaved thin fluviatile and lacustrine sediments occur.	Rapidly subsiding basin supporting standing bodies of water.
—unconformity—		
Ketilidian basement—including the Julianehåb granite		

magmatism incorporates the lavas of the Eriksfjord Formation; the younger magmatism is entirely intrusive in nature.

The lavas of the Eriksfjord Formation are the subject of this study. The majority of flows have basaltic to hawaiitic compositions (Upton and Emeleus, 1987). Three major lava groups are recognised, each one separated by thick units of continental sandstones (Table 2). The Eriksfjord Formation predates mid-Gardar WNW dykes and the South Qôroq intrusion (Table 1). Its relationship to the early Gardar plutons is unclear but it may predate the North Qôroq intrusion and therefore be about 1300 Ma. On general geological grounds it is inferred to be one of the earliest events in the evolution of the province. The lower (Mussartût) group shows a change from initially alkaline ultramafic magmatism, associated with explosive activity, to olivine-deficient, glassy basalts in later Mussartût flows. The middle (Ulukasik) lavas, however, are more porphyritic, mildly alkaline olivine basalts and hawaiites. The absence of pyroclastics and the pahoehoe nature of the Ulukasik flows suggest that they erupted from low-angle shield volcanoes or fissures. The upper (Ilímaussaq) group is the thickest (ca. 1100 m) and most petrologically-varied group. The lowest flows are "star basalts", so named because of the presence of radiating aggregates of white plagioclase phenocrysts. These pass into aphyric flows, "platy basalts" (feldspar–phyric with plagioclase megacrysts up to 20 cm long) and trachytic flows at the top of the pile.

The absence of interleaved sediments or appreciable erosion of lava surfaces implies that lava eruption rates were high. Hence, although accurate dates are unavailable, the whole of the Eriksfjord Formation may well have formed

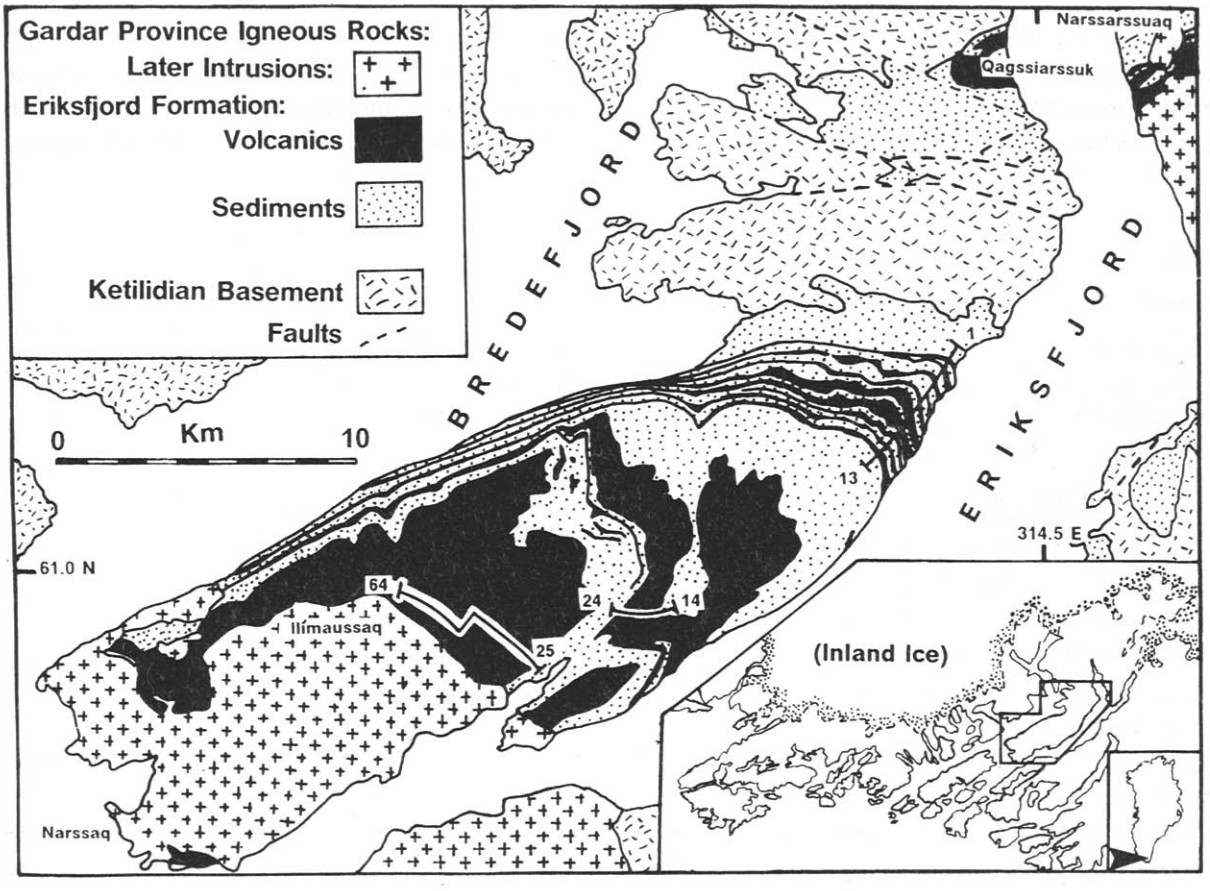


Fig. 1. Sketch geological map showing sampling sites for this study. (Redrawn from Poulsen, 1964.)

within a period of 5–10 Ma (B.G.J. Upton, pers. commun., 1990). All lava flows have experienced low grade hydrothermal (300–450 °C) metamorphism, with epidote well developed. The sequence is unaffected by later orogenic events and is essentially undeformed and unmetamorphosed save for the regional influence of the hydrothermal activity and local effects in the vicinity of the younger (ca. 1140 Ma) Ilímaussaq alkaline intrusion (Fig. 1).

Sixty-four lava flows were sampled in the three stratigraphic units (Fig. 1). These sections cover the entire 1500 m of lava stratigraphy and comprise thirteen flows from the lower group, eleven flows from the middle group and forty flows from the upper group. Between five and ten three-inch cores were drilled from each flow using a portable two-stroke motor and water-cooled diamond drill bit. The cores were oriented in situ, before removal from the outcrop, using a sun compass in conjunction with a magnetic compass. Four one-inch cores were cut from each site in the laboratory, for palaeomagnetic measurements.

3. Laboratory techniques

Piper (1977) used A.F. demagnetisation in his original study, and stereographic projections only to analyse the data. The present study incorporates stepwise thermal demagnetisation of all samples using a Magnetic Measurements Thermal Demagnetiser. Magnetisations were measured on a Molspin spinner magnetometer. All demagnetisation and measurement took place inside a 2-m^3 Reubens coil arrangement, designed to minimise the effect of the ambient magnetic field in the laboratory. During the experiments, fluxgate magnetometer measurements showed the field to be as low as 0.01 μT inside the oven and 0.1 μT inside the cage.

A pilot demagnetisation study of one sample per site (i.e. one per flow) was performed in 50 ° or 100 ° steps up to 500 °C, then in 10 ° or 20 ° steps to the maximum unblocking temperature, to isolate the blocking temperature spectra for each flow. Subsequently, four samples per site were thermally demagnetised and measured, using the appropriate temperature steps defined by the pilot study. The results were analysed using orthogonal vector plots (Zijderveld, 1967) in conjunction with plots of normalised intensity (J/J_0) with temperature and stereographic projections to isolate the components of magnetisation present in the samples. Tilt-corrected site mean directions and palaeomagnetic poles were calculated using computer programs incorporating Fisher statistics (Fisher, 1953).

4. Thermal demagnetisation results

Table 3 shows the results of the thermal demagnetisation study. The remanence of each sample is categorised (A–E) according to the number of components isolated by vector analysis (see Table 3 legend). Approximately half of the lava flows (52%) in the upper (Ilímaussaq) group possess a single component of magnetisation. Figure 2 (a, b) shows orthogonal vector plots of typical normal and reversed flows, respectively, from the upper lava group. They indicate the single-component nature of the majority of flows within the upper group. It is only in the top and, to a certain extent, the middle of the group that two and, very rarely three, components are isolated (Fig. 2c). The top of the lava pile lies adjacent to the ca. 1140 Ma Ilímaussaq intrusion (Fig. 1). Hence, the multi-component nature of the magnetisations in this section is possibly due to the magnetic overprinting effect of this intrusion. Figure 2 shows the stable directions of magnetisation obtained from a flow close to the top of the pile (Fig. 2d) and from a marginal syenite of the Ilímaussaq intrusion (Fig. 2e). Clearly, these two directions are comparable with each other and different from the directions seen in the other flows. Some flows show a partially overprinted direction. Unlike the totally overprinted flows, the primary remanence can still be recovered from such samples. All flows which show the totally overprinted direction as the stable remanence are omitted from the calculation of the group mean direction for the upper lavas.

Figure 3 shows typical primary components of magnetisation for samples from the middle (Ulukasik) and lower (Mussartût) groups. The majority of flows from both groups possess single,

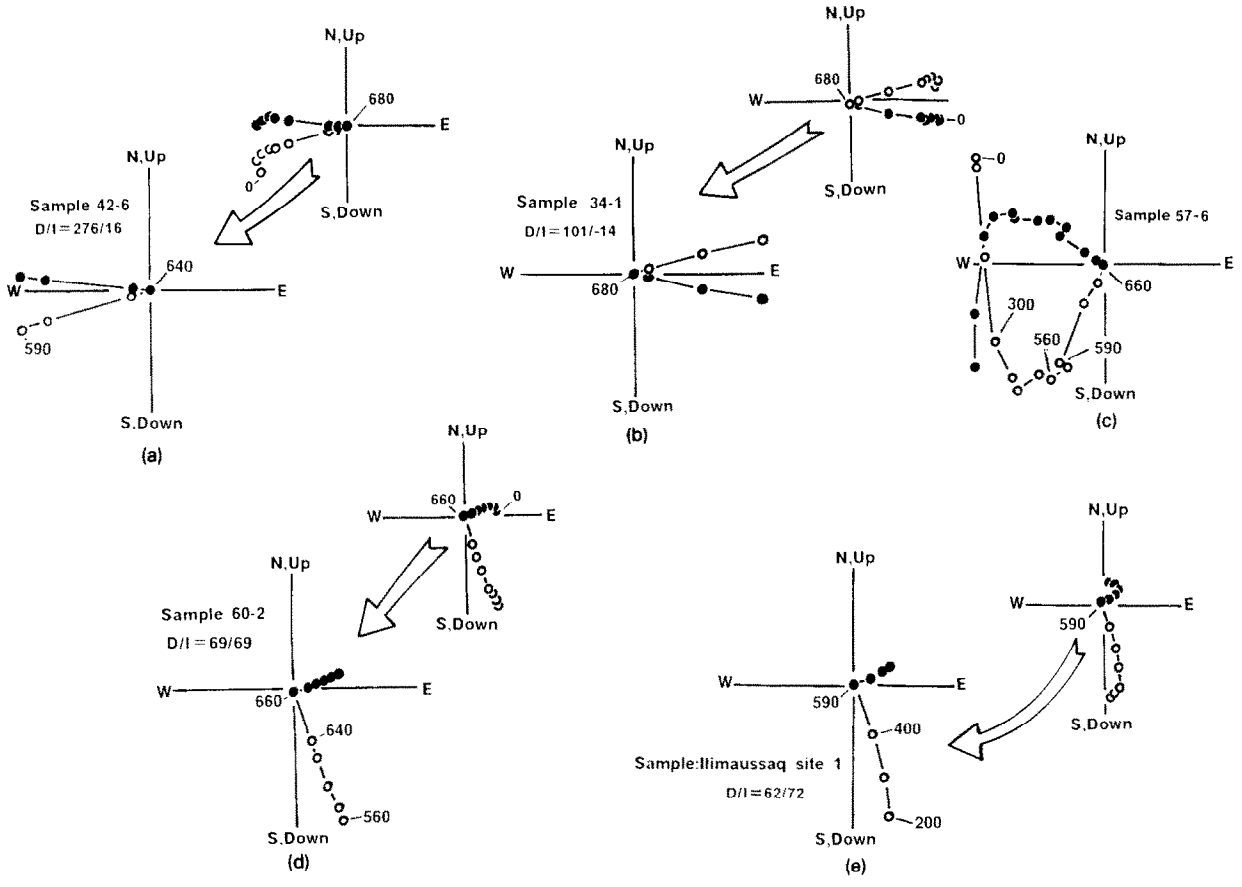


Fig. 2. Orthogonal vector (Zijderveld) plots showing the primary component of magnetisation for lava flows from the upper lava group. Unit of J is $\times 10^{-5}$ A m^2 kg^{-1}. (a) Normally magnetised flow. (b) Reversely magnetised flow. (c) Flow showing a multi-component remanence. (d) Flow showing overprinted direction. (e) Stable remanence direction from a marginal syenite of the Ilímaussaq intrusion.

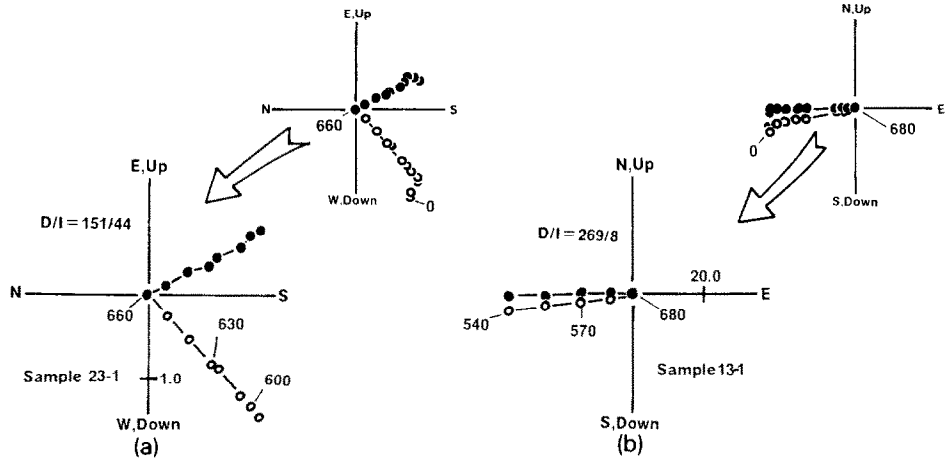


Fig. 3. (a) Orthogonal vector (Zijderveld) plot showing a typical primary component of magnetisation for the middle lava group. (b) Orthogonal vector (Zijderveld) plot showing a typical primary component of magnetisation for the lower lava group.

TABLE 3

Results of thermal demagnetisation and component analysis

Site	*N*	*D*	*I*	*T* (1°)	Polarity	Category	Rock mag.
Upper lavas:							
064	4	286	23	500–590	N	B	MD/SD
063	4	261	56	400–660	N	B	MD/SD
062*	4	270	83	530–640	I	C	TiH
061	4	293	60	600–680	N	C	MDM
060*	4	338	76	530–680	I	C	TiH
059*	4	344	80	575–680	I	C	MDM?
058*	4	324	76	400–680	I	C	TiM-M
057	4	270	56	400–575	N	C	TiM-M
056	4	278	40	575–680	N	E	TiH?
055	4	290	58	575–680	N	C	MD/SD
054	4	319	46	575–680	N	C	TiM-M
053*	4	20	62	540–630	I	C	TiM-M
052	4	307	44	400–590	N	C	MD/SD
051*	4	333	51	540–615	I	B	H
050*	4	264	73	400–590	I	B	TiM-M
049*	4	325	36	400–590	I	B	H
048*	4	345	51	400–560	I	B	MD/SD
047*	4	326	44	500–640	I	B	H
046	4	120	24	300–560	R?	A	MDM
045	4	89	62	400–560	R?	C	MDM
044	4	278	30	500–620	N	B	TiM-M
043	4	276	23	300–620	N	A	H
042	4	280	23	530–640	N	A	H
041	4	299	21	530–640	N	C	H
040*	4	309	75	300–560	I	A	MDM
039	4	102	1	200–530	R	C	MD/SD
038	4	102	−12	560–640	R	B	TiM-M
037	4	102	−28	500–640	R	A	H
036	4	99	−23	500–670	R	A	H
035	4	102	−35	560–670	R	A	H
034	4	98	−16	530–680	R	A	H
033	4	91	−18	620–680	R	B	H
032	4	281	52	500–590	N	B	TiH
031	4	305	31	590–680	N	B	H
030	4	318	31	620–680	N	D	H
029	4	311	27	500–640	N	B	H
028	4	314	50	500–560	N	A	MDM
027	4	312	25	590–680	N	B	TiM-M
026	4	307	33	400–590	N	B	MD/SD
025	4	311	56	300–560	N	B	MD/SD
Middle lavas:							
024*	4	296	43	600–700	N	B	H
023*	4	314	1	600–700	N	B	H
022*	4	103	39	400–560	I	B	MD/SD
021	4	154	−3	400–600	I	B	H
020	4	150	41	580–700	I	B/C	H
019	4	157	47	530–680	I	B	H
018	4	155	44	590–700	I	B	H
017	4	154	50	580–700	I	B	H

TABLE 3 (continued)

Site	N	D	I	T (1°)	Polarity	Category	Rock mag.
Middle lavas:							
016	4	151	48	560–700	I	B	H
015	4	150	43	560–700	I	B	H
014	4	160	52	300–590	I	B	MD/SD
Lower lavas:							
013*	3	94	−9	540–660	R	B	MDM
013	4	291	8	500–580	N	A	MD/SD
012	4	287	16	400–620	N	A	MD/SD
011	4	296	24	500–580	N	B	MDM
010	4	297	31	400–560	N	B	MDM
009	4	295	14	500–580	N	B	MDM
008*	4	89	3	500–580	R?	B/A	MDM
007	4	256	36	500–580	N	B	MD/SD
006	4	295	−5	400–580	N	B	MDM
005	4	292	27	400–600	N	A/B	MD/SD
004*	4	24	9	500–560	I	B	MD/SD
003	4	326	34	400–600	N	A	MDM
002	4	302	21	500–560	N	B	MD/SD
001*	4	355	56	300–530	I	B	MD/SD
001A*	3	108	−39	200–560	R	B	MD/SD

N = number of samples per site; D, I = declination and inclination of primary component; T (1°) = temperature interval defined by primary component.

* = Samples omitted from calculation of mean directions.

Categories of remanence: A = single high-temperature component; B = single high-temperature component with a low-temperature viscous; C = two components (one high-temperature and one low-temperature), no viscous; D = two components and viscous; E = more than two components.

MDM = multi-domain magnetite only. MD/SD = distributed grain size of magnetite with multi-domain predominating over single or pseudo-single domain; one magnetic mineral phase. TiM-MDM = two magnetic mineral phases—Ti-rich titanomagnetite and multi-domain pure magnetite. TiH = titanohaematite. H = haematite.

non-viscous components. The flows from the lower group show the greatest stability to thermal demagnetisation of the entire collection.

Figure 4 shows the different types of curves for normalised intensity (J/J_0), as a function of demagnetisation temperature, seen in the collection. Fifty percent of flows in the upper group have distributed blocking temperature (T_b) spectra (Fig. 4, a and b) and fifty percent have narrow spectra with high blocking temperatures (Fig. 4, c and d). Samples which have been overprinted show no preferential curve type. In the middle group, eight of eleven flows have "square shaped" J/J_0 vs. T curves (Fig. 4d). These correspond to flows showing an anomalous SE positive direction, contrasting with the easterly negative/westerly positive axis defined by the remainder of the collection. These sites all show a single, high blocking temperature component with a narrow blocking temperature range. Twelve of thirteen flows from the lower group show the curves seen in Fig. 4 (c, d). In extreme cases, up to 90% of the remanence is lost in a 20–30 °C interval, just below the maximum unblocking temperature. Only two flows, both in the top of the upper group, show distributed T_b spectra with 80% lost before 200 °C.

The site mean directions for each lava group are shown in Fig. 5. The upper lavas show two populations of normal (westerly) directions—one with shallow to intermediate (0–60°) inclinations and W to WNW declinations and the other with steep (> 70°) inclinations and close to the present earth field (P.E.F.) direction at this locality

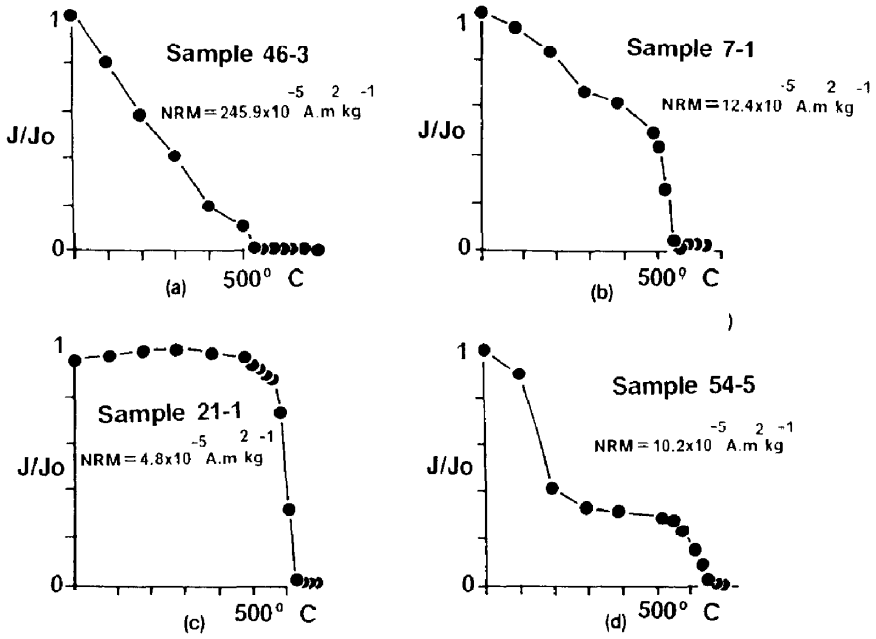


Fig. 4. Graphs showing the four types of behaviour of normalised intensity (J/J_0) against demagnetisation temperature observed in this study. Units of J are $\times 10^{-5}$ A m^2 kg^{-1}.

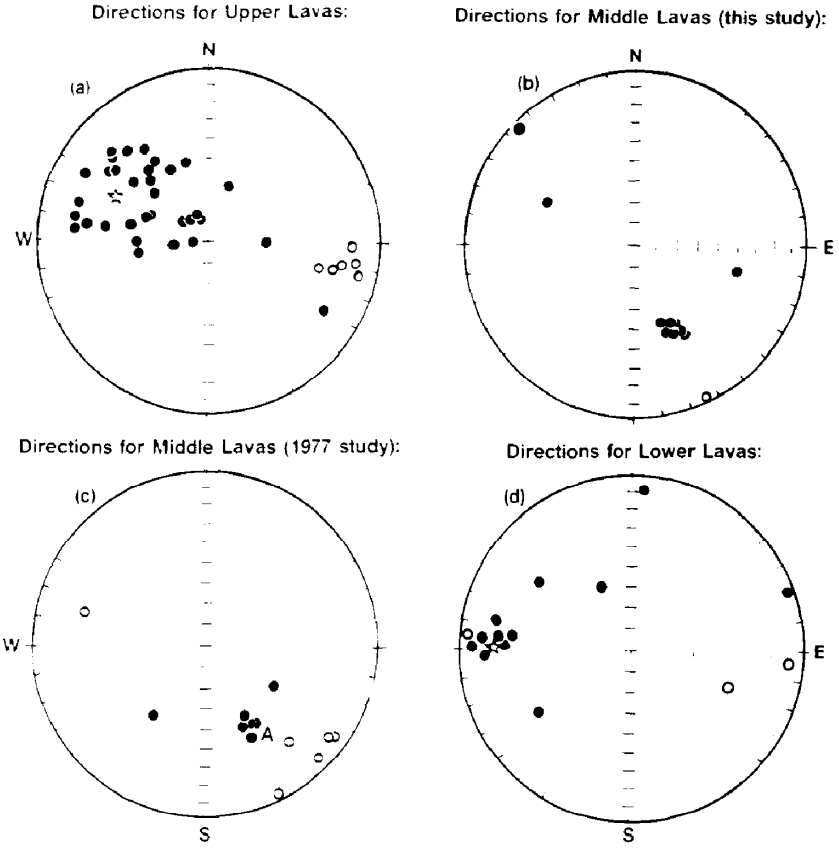


Fig. 5. Site mean directions calculated in this study. (a) Upper lava group. (b) Lower lava group. (c) Middle lava group (this study). (d) Middle lava group (results of Piper, 1977).

($I = 78°$). It is therefore plausible that these sites have become totally remagnetised in the P.E.F. Alternatively, they may have been overprinted by a late Gardar igneous event, such as the Ilímaussaq, which shows a steep E to NE direction compatible with this group (Piper, 1976). The reversed (easterly) directions are much more closely grouped than the normal directions, in a shallow E to ESE direction.

The site mean directions for the middle lava group (Fig. 5b) show only one population as opposed to the two found by Piper (1977). In that study, the two mean directions calculated for the middle group differed by 15.5° in declination and 59.2° in inclination (Table 4). In this study, however, the site mean direction for the middle lava group ($D = 153.8°$, $I = 46.1°$, $\alpha_{95} = 3.6°$) is very well defined and is almost identical to Piper's A direction ($D = 151.0°$, $I = 46.4°$, $\alpha_{95} = 4.3°$). No evidence is found for Piper's B direction, which is considered to be in error, although the source of error is not obvious. As can be seen from Fig. 5b, the grouping of site means about the B direction is poor. It is possible that a systematic field measurement (orientation) error may be the cause of these spurious directions but this is unclear. The mean direction for the middle group calculated in this study bears no similarity with either the normal or reversed directions from either the upper or lower groups. The same observation was found by Piper (1977). The geomagnetic significance of this anomalous direction is discussed in Section 9.

The top and bottom flows of the lower group have reversed (easterly-negative) directions. Both reversals, R–N at the base and N–R at the top are symmetrical, in contrast to those found in the upper group. The remaining flows in the lower

TABLE 4

Site mean directions and rotated pole positions for the Gardar lavas

Lava group	Study	Mean direction		Rotated pole position		Statistics		
		D	I	Lat.	Long.	N	α_{95}	k
Upper lava group:								
Normal	This study	295.5	40.5	37.6N	198.9E	20	7.9	18.2
Combined (normal and reversed)								
	Piper (1977)	279.5	24.0	20.3N	206.3E	24*	8.4	13.0
Upper lava group:								
Reversed	This study	97.4	−14.5	14.7N	205.1E	7	9.3	43.2
	Piper (1977)	–	–	–	–	–	–	–
Middle lava group:								
Intermediate	This study	153.8	46.1	4.4S	319.9E	7	3.6	286.0
	Piper (1977)							
	A	151.0	46.4	3.7S	322.7E	5	4.3	317.0
	B**	135.5	−12.8	32.3S	348.4E	5	13.9	31.0
Lower lava group:								
Normal	This study	270.4	21.6	14.2N	213.3E	10	11.9	17.4
	Piper (1977)	260.0	25.0	10.5N	222.8E	7	13.1	22.0

D, I = Mean declination and inclination values; N = number of sites; α_{95} = semi-angle of 95% confidence cone; k = precision parameter.

* Calculation of pole position includes 12 reversed flows in addition to 12 normal flows (combined normal and reversed pole).

** Anomalous data. Pole B is thought to be in error.

*** Rotated poles calculated after closure of the Davis Strait by a combined rotation of Greenland to Europe and Europe/Greenland to North America ammounting to 18° clockwise rotation about an Euler pole at 265.6E, 70.5N, according to the Bullard et al. (1965) fit.

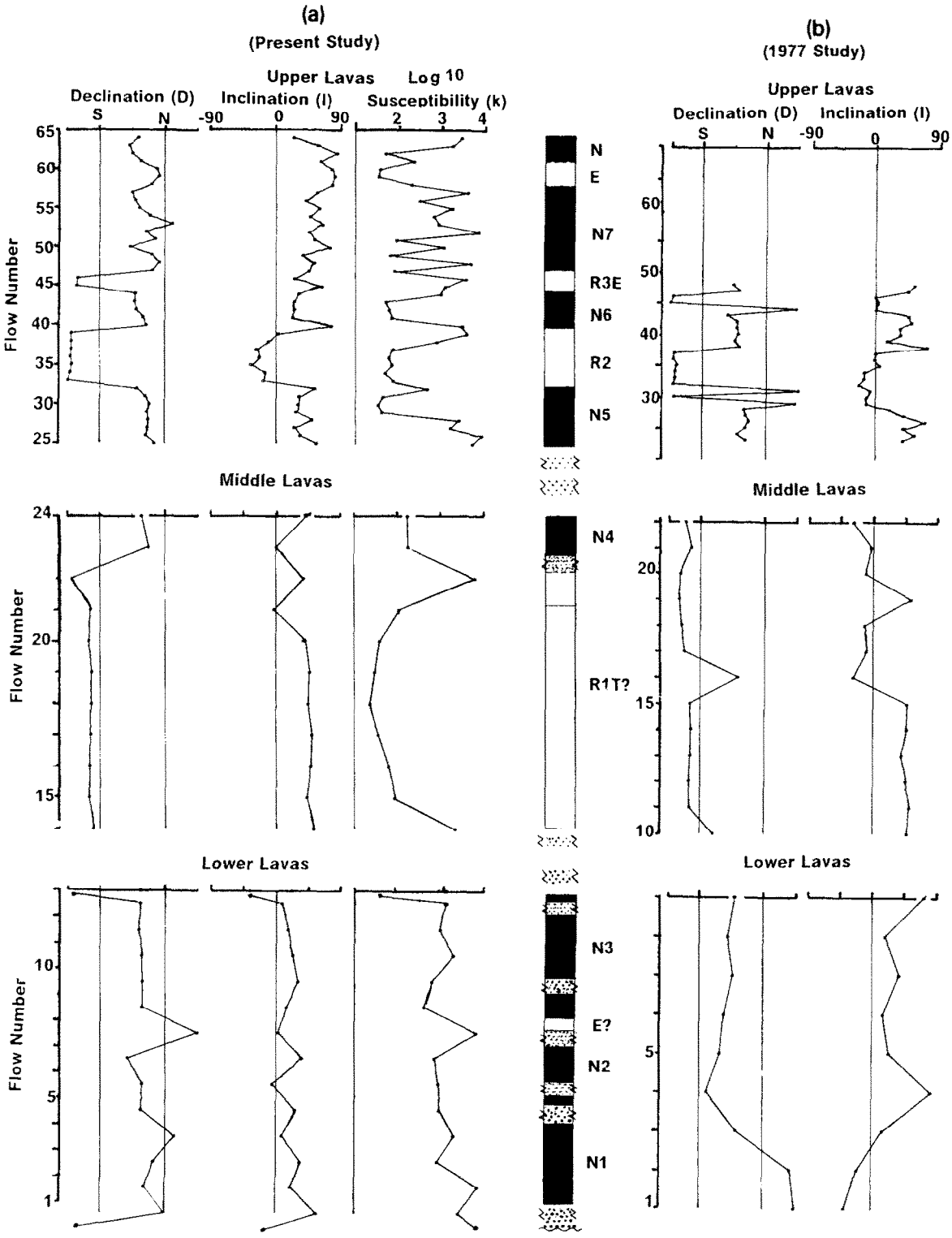


Fig. 6. Comparison of the two magnetostratigraphy diagrams for the Gardar lavas: (a) results from this study; (b) results from Piper (1977). Each point on the diagrams is a flow mean primary direction. Vertical scale figures indicate the flow number. N = normal; R = reversed; T = transition?; E = excursion?.

group show normal (westerly-positive) directions, with the exception of a single flow (Figs. 5c, 6b). Magnetic anisotropy experiments show that the direction exhibited by this flow is not due to the presence of a magnetic fabric within the flow (Thomas, 1991). The existence of a thin layer (~ 4 m) of sediment between this flow and the one immediately below it is indicative of a sizeable time gap between the eruption of these two flows, probably enough to record an excursion in the geomagnetic field. This flow is excluded from the calculation of the group mean direction.

5. Magnetostratigraphy

The Eriksfjord Formation is one of the very few geological settings of Proterozoic age which offer a suitable section for defining a polarity stratigraphy; continuous lava sequences with interleaved sediments having a total thickness of 3 km are otherwise rare. The magnetic stratigraphy defined by Piper (1977), shown in Fig. 6b, covers only the bottom two thirds of the lava stratigraphy, i.e. the whole of the lower and middle lava groups and part of the upper group, up to and including the "platy basalt" flows. There are some 180–200 m, comprising nineteen flows, above the "platy basalts". Results from these nineteen flows, in addition to the remainder of the lava pile, together totalling 66 flows, define the new magnetostratigraphy shown in Fig. 6a. The two plots show critical differences. In the upper lava group, four complete reversals are observed in the previous study, whereas only two complete reversals are identified by the present study. Two flows, within the division of "platy basalt" flows, record easterly directions, which appear to correlate with the upper two reversals in the previous study (Fig. 6). In the new study, these flows record directions which show characteristic reversed (easterly) declinations but the inclinations are positive, hence no complete reversal has been recognised.

There are no further reversals seen in the top part of the upper group, which was not sampled by Piper (1977). Figure 6a shows that in this section, certain flows exhibit directions close to the P.E.F. direction and others are contaminated by the magnetic effect of the Ilímaussaq intrusion. It is difficult to separate these two possible explanations because it appears that the Ilímaussaq intrusion cooled in a field direction close to the P.E.F. (Fig. 2e). The inclination values seen for flows in this top part of the succession are steeper than those for flows below the "platy basalts".

The middle (Ulukasik) lava group shows an anomalous direction which bears no resemblance to those from the upper and lower lava groups. The group shows a southeasterly intermediate positive mean direction. Only at the top of the group, where the lava flows are interleaved with thin sediments, are directions of normal polarity, similar to the rest of the succession observed. The polarity change is unrecorded but possibly occurs within the sediments unsampled for this study. Comparison of Figs. 6a and 6b shows that only a single population of directions was found, as opposed to the two (A and B) populations found by Piper (1977).

The results from the lower (Mussartût) lava group provide a limited temporal coverage; the total thickness of interleaved lava flows, sediments and conglomerates which form the group is some 740 m, of which the lavas contribute only some 400 m (Table 2). At first glance, the Mussartût unit appears to reflect a constant normal polarity interval. This may be misleading. The lack of results from the 300 m of interleaved sandstones means that the magnetostratigraphy from this unit is incomplete. The polarity stratigraphy for the lower flows is much better defined than in the previous study, where, at the base of the Mussartût group, in a formation thickness of some 200 m only two lava flows were sampled and only nine were sampled in the whole of the group, as opposed to the fifteen in this study. One or more polarity reversals, between the lower and middle lava groups, may be recorded in the quartzitic sediments of the Naujarssuit (middle) sandstone group, which were not sampled.

6. Pole positions and apparent polar wander

Four new pole positions were calculated for the Gardar lavas: two for the upper lava group

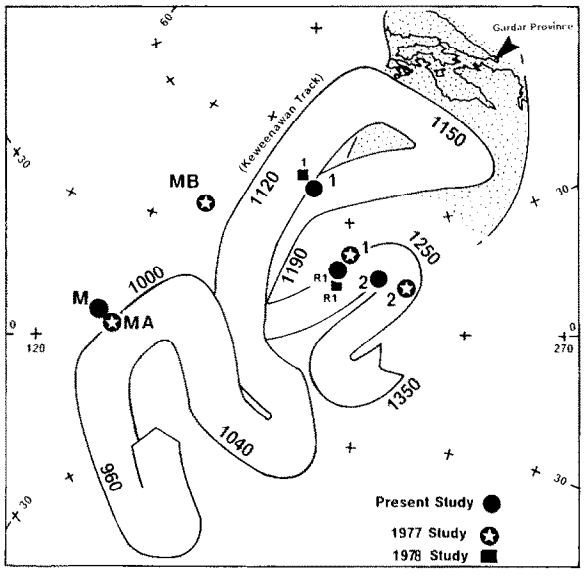


Fig. 7. Sketch map showing position of the palaeomagnetic pole positions for the Gardar lavas on the Proterozoic APWP for the Laurentian craton after the closure of the Davis Strait by rotating Greenland and Europe back to the North American craton according to the Bullard et al. (1965) fit of the North Atlantic continents. Poles: *1* = upper lava group, combined pole (Piper, 1977); *MA* = middle lava group A pole (Piper, 1977); *MB* = middle lava group B pole (Piper, 1977); *2* = lower lava group (Piper, 1977). 1 = upper lava group normal pole (Pesonen, 1978); *R1* = upper lava group reversed pole (Pesonen, 1978). *1* = upper lava group normal pole (this study); *R1* = upper lava group reversed pole (this study); *M* = middle lava group pole (this study); *2* = lower lava group pole (this study).

and one each for the middle and lower groups (Table 4). The upper group was split into normal and reversed flows and one pole position calculated for each polarity. This approach follows that of Pesonen (1978) and differs from that of Piper (1977) who calculated a single pole position for the upper lavas from a mean magnetisation direction derived from flows of both polarities. Consequently, the Piper pole from the upper group differs considerably from the pole calculated for the normally magnetised upper lavas in this study (Fig. 7, Table 4). This has significant implications for the relative positions of the two poles on the contemporaneous apparent polar wander path (APWP) for the Laurentian craton incorporating the Logan and Keweenawan loops, shown in Fig. 7. The new poles were plotted on the APWP after reconstruction of the Laurentian shield by closing the Davis Strait and using a combined rotation of Greenland to Europe and Europe and Greenland to North America, as described by Bullard et al. (1965). Three sets of poles are shown for the Gardar lavas on Fig. 7: those of Piper (1977), the reinterpretation of that data by Pesonen (1978) and the new data. The similarity between Pesonen's recalculated pole for the upper lavas normal flows (ULN) and the new ULN pole is striking. These differ markedly from the original Piper UL pole. This pole plots on the 1190–1250 Ma section of the Logan Loop, whereas the other two plot on the 1120–1150 Ma section, a time interval appreciably younger than the inferred age of the Eriksfjord Formation (Table 1).

The pole positions for the middle lava group do not coincide with any part of the APWP and are clearly anomalous. The pole position for the lower lavas also plots on a younger section of the APWP than the corresponding pole given by Piper (1977).

7. Rock magnetic studies

An extensive rock magnetic study of samples from all 66 flows from the Gardar lava succession was performed. Measurements of Curie temperatures, temperature-dependent susceptibility, anisotropy of magnetic susceptibility and hysteresis parameters yielded results summarised in Table 3. A detailed account of this work will be published elsewhere (Thomas, 1991).

86% of all flows have a single magnetic mineral phase (Fig. 8). Magnetite (42.6% of flows) and haematite (31%) are the predominant remanence carriers. High and low temperature susceptibility and hysteresis results indicate that the dominant magnetite grain size is multi-domain (MD). 13% of flows carry mainly MD magnetite, many flows (35%) have a distributed grain size of magnetite, with MD always dominating over SD/PSD. 14% of flows, all in the upper group, appear to possess two magnetic mineral phases—a pure magnetite ($T_c = 580\,^\circ$C) and possibly a titanomagnetite phase (TM30, $T_c = 430\,^\circ$C). However, at present this is unclear and needs further investigation. Significantly, the flows

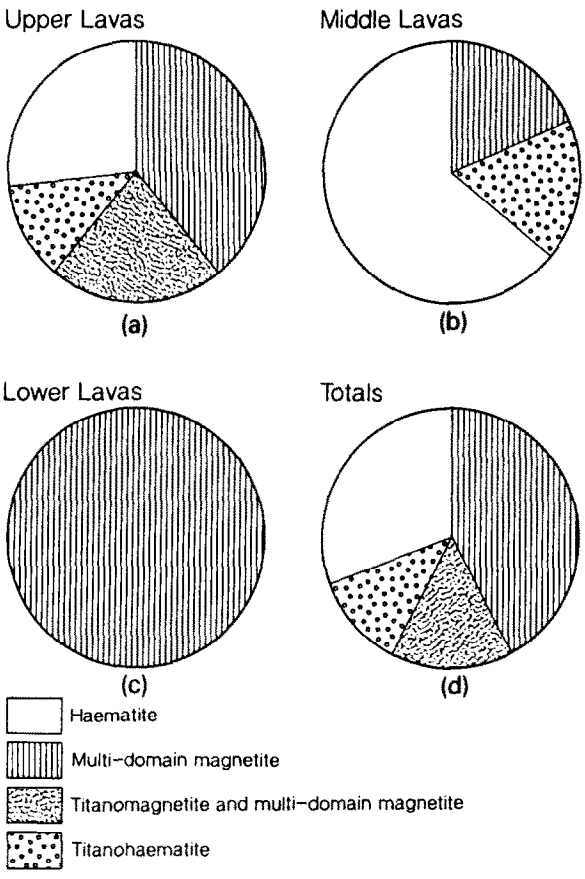


Fig. 8. Pie charts summarising the findings of the rock magnetic study of the Gardar lavas (Thomas, 1991). Segments indicate the number of flows in each lava group which carry the following magnetic minerals: multi-domain magnetite; titanomagnetite(?) and multi-domain magnetite; titanohaematite; haematite.

thought to contain two mineral phases are located in the middle to top of the upper lava group and correspond to flows possessing two or more non-viscous components of magnetisation (categories B and C). This is further evidence supporting the suggestion that part of the upper lava group has been partially overprinted.

The magnetite-bearing flows are predominantly of normal polarity and the haematite-bearing flows are mostly reversed. This concurs with the observation of Piper (1977) that flows with easterly directions have higher oxidation states than those with westerly directions. The significance of this polarity–mineralogy correlation is uncertain but a similar correlation between oxidation state and polarity has been recognised in Tertiary lava successions (e.g. Adc-Hall and Wilson, 1969).

8. Geomagnetic field model

The presence of asymmetric reversals in Upper Proterozoic rocks is not uncommon. Palmer (1970) and Pesonen (1978) found one, and possibly as many as three successive asymmetric reversals (R → N → R → N) within lava successions from the Keweenawan igneous province (ca. 1150–1100 Ma). These reversals have been modelled using both inclination asymmetries and reversed/normal palaeointensity ratios (Nevanlinna and Pesonen, 1983). Palaeointensity studies on the Gardar lavas are, at present, incomplete. Thus, the field model proposed here is based on results from the observed inclination asymmetry only.

Given a colatitude (θ), then:

$$\theta = 90 - \lambda \tag{1}$$

$$\tan I' = \left[2\cos\theta + g_2^0/g_1^0(9/2\cos^2\theta - 3/2) + g_3^0/g_1^0(10\cos^3\theta - 6\cos\theta)\right] \times \left[\sin\theta + g_2^0/g_1^0(3\cos\theta\sin\theta) + g_3^0/g_1^0(15/2\cos^2\theta\sin\theta - 3\sin\theta)\right]^{-1} \tag{2}$$

$$x/a = g_2^0/2g_1^0 \tag{3}$$

I' = observed inclination; a = radius of the earth.

The colatitude (θ) as calculated from eqn. (1) is 75.4°, where λ is the palaeolatitude of the upper lava group, calculated from the mean of the normal and reversed inclinations. Assuming the inclination asymmetry to be due to a reversing geocentric dipole term (g_1^0) and a non-reversing quadrupole term (g_2^0) and neglecting octupole terms (g_3^0), then eqn. (2) (Merrill and McElhinny, 1983) can be simplified. The ratio of quadrupole to dipole terms (g_2^0/g_1^0) was changed iteratively until the observed normal and reversed inclination values were satisfied. This value of g_2^0/g_1^0 was used in eqn. (3) which gave an equivalent dipole offset, x, of 570 km northwards. For the new data set of the Gardar lavas, $g_2^0/g_1^0 = 0.18$

satisfies the inclination anomaly observed, i.e. the quadrupole term was 18% of the dipole term.

This non-geocentric axial dipole field model may explain the discrepancy between the predicted and observed positions of the poles on the APWP path for Laurentia. Since the axial geocentric dipole model is clearly violated, the use of the APWP for dating these poles is not valid.

9. Discussion and conclusions

The new magnetostratigraphy is a more complete record of the behaviour of the geomagnetic field during Gardar times than previously presented. The new data suggest that the reversal frequency during the eruption of the Gardar lavas may not be as high as previously suspected (Piper, 1977). Two complete reversals (N → R → N) are found in the upper (Ilímaussaq) group as opposed to the four identified by Piper (1977). The thermal demagnetisation method was unable to completely isolate the two reversals seen in the "platy basalts" in the previous, A.F. demagnetisation, study. In the top 200 m of the lava pile, no reversals were recorded but periodic shifts in direction are seen. This is unlikely to represent variation in the ancient field when the lavas were erupted, because the component structure shows that the characteristic lava remanence in this part of the upper group is first partially and then totally overprinted as the invading Ilímaussaq intrusion is approached.

Both complete reversals recorded show inclination asymmetries of 30°, which support the findings of Piper (1977). Clearly, directional data are needed from the continental sandstones interleaved with the lava flows in order to fully test the reversal asymmetry throughout the entire Eriksfjord Formation. The geomagnetic field model proposed to account for the inclination asymmetry in the new data is an offset dipole model, with 570 km of axial (northward) displacement. The accuracy of this model can only be tested by the comparison of palaeointensity results from normal and reversely magnetised flows, as shown by Nevanlinna and Pesonen (1983). Thus, a more complex model is not precluded by the present evidence.

Results from the middle (Ulukasik) lava group continue to be enigmatic and are the major discussion point arising from this work. The issue is simplified by the existence of only a single population of directions but the question remains as to why this single mean direction appears to be unrepresentative of either the normal or reversed directions within the rest of the lava succession. There may be several possible explanations. One is that regional hydrothermal alteration has caused total remagnetisation of the middle lava group. The palaeomagnetic evidence from both this new study and the previous work suggest that this explanation is unlikely. In both studies, all samples from the middle lava group show single components of magnetisation with high blocking temperatures (> 600 °C). Geological evidence estimates hydrothermal temperatures of 300–450 °C. These temperatures are too low to have unblocked a TRM component possessing such a high blocking temperature (Pullaiah et al., 1975) over the likely duration (up to several millions of years) of the hydrothermal event. Moreover, there is no obvious reason why *only* the middle of the three groups should be remagnetised as a result of regional hydrothermal alteration, to produce a secondary chemical remanence (CRM). The high oxidation states of flows from the middle lavas (Piper, 1977) suggests that deuteric oxidation may be responsible for the haematite present.

Piper (1977) attributed the anomalous direction to either an apparent polar wander excursion or a transitional field direction. He favoured the former since the higher lavas in the middle group are separated by sandstones and their eruption must have scanned a considerable period of time. However, the new results question this interpretation. Figure 6a shows that a directional change occurs at the top of the middle lava group, with the change lost in the thin sediments between two flows. Crucially, the direction after the change is similar to the normal polarity (westerly) directions witnessed in the upper and lower lava groups. This change to a normal direction is not seen in Piper's results (Fig. 6b). A similar change, unrecorded by this study, occurs between the middle and lower groups. This evidence suggests that the anomalous middle lava direction could

indeed represent a transitional field direction. Geological evidence also supports this interpretation. The absence of vesicular flow tops and intervening clastics in most of the group, in addition to the presence of pahoehoe surfaces indicate a lack of erosion and a small time gap between eruption of successive flows. The whole of the middle group (including the sedimentary layers at the top) is only 250 m thick. This undoubtedly constitutes the shortest eruption time period of all three lava groups—probably of the order of a few hundred years, or less. It is therefore conceivable that during this short time period, the geomagnetic field could have been locked in a transitional phase. It is well documented (e.g. Coe, 1967; Dagley and Wilson, 1971; Bogue and Coe, 1981) that the field intensity decreases substantially during a field reversal. Thus, the calculation of palaeointensity values for the middle lavas and palaeomagnetic analysis of the sandstone units containing the polarity changes are needed to test the proposition that the middle lavas were erupted during a polarity transition.

The polarity stratigraphy for the lower (Mussartût) unit is incomplete due to the lack of data from the sandstone units interleaved with the lava flows. Two reversals were found, one of which was identified in the previous study. A further flow records an easterly direction. This direction is not due to the effect of a dominant magnetic fabric (Thomas, 1991), hence it must be considered to be real and is tentatively interpreted as being a record of a brief palaeoexcursion of the geomagnetic field.

Geochemical evidence (Upton and Emeleus, 1987) indicates a generalised increase in differentiation from the lower to the upper lava group, suggesting that the lava source may have been a single magma chamber, or a closely related nest of chambers, which cooled and evolved compositionally with time (B.G.J. Upton, pers. commun., 1990). This is important for estimating the time span of the geomagnetic changes identified here, because single volcanic systems rarely have lifespans in excess of 5 Ma. This implies that the entire Eriksfjord Formation was created within this time, and the Gardar lavas themselves, in about 0.75–1 Ma. Thus, the magnetostratigraphy documented here records the behaviour of the geomagnetic field over only a very short interval of geological time. During this short time, the field direction changed rapidly in the upper lava group, where no transitional directions are recognised, but remained locked in at a transitional direction for a longer period during the eruption of the middle lava group. This illustrates some analogies with the behaviour recognised in the Miocene Steens Mountain lavas (Prévot et al., 1985).

Results from both this study and rock magnetic studies of the Gardar lavas (Thomas, 1991) suggest that the existence of pure magnetite and haematite, and the absence of titanium-rich titanomagnetites in the lavas may result from various stages of high temperature (deuteric) oxidation. Palaeointensity studies, ^{39}Ar:^{40}Ar and U–Pb dating (B. Paterson, pers. commun., 1990) currently in progress will constrain the cause and timing of the geomagnetic features documented here.

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